Design of IIR filter (2)

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CHAPTER

Infinite Impulse Response Filter Design

8

Reference book
Chapter 8: Infinite Impulse Response Filter Design
8.2 Bilinear Transformation Design Method
(Page 303-317)

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Discrete time system v.s. continuous time system

Discrete time system

Transfer function of a discrete time system:

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{i=0}^{N} a_i \, z^{-i}}{1 + \sum_{i=1}^{N} b_i \, z^{-i}}$$

Continuous time system

Transfer function of a continuous time system

$$H(s) = \frac{Y(s)}{X(s)} = \frac{\sum_{i=0}^{N} a_i \, s^{i}}{1 + \sum_{i=1}^{N} b_i \, s^{i}}$$

$X()$ represents the input signal,
$Y()$ represents the output signal

Two categories of discrete-time filters

Digital filters are divided into two categories according to their impulse response:

Infinite Impulse Response filters (IIR filters)

$$y(n) = \sum_{i=0}^{N} a_i\, x(n-i) - \sum_{i=1}^{N} b_i\, y(n-i)$$

$$H(z) = \frac{\sum_{i=0}^{N} a_i\, z^{-i}}{1 + \sum_{i=1}^{N} b_i\, z^{-i}}$$

Finite Impulse Response filters (FIR filters)

$$y(n) = \sum_{i=0}^{N} a_i\, x(n-i)$$

$$H(z) = \sum_{i=0}^{N} a_i\, z^{-i}$$

Recap: Design of digital IIR filter

1. The filter's specifications are drawn up
2. The z-domain transfer function of the filter is set up using

Matched z-transform:
Transfer poles and zeros from s-domain to z-domain

$$H(s) = \frac{(s - z_1)(s - z_2) \dots (s - z_N)}{(s - p_1)(s - p_2) \dots (s - p_N)}$$

$$H(z) = \frac{(z - e^{z_1 T})(z - e^{z_2 T}) \dots (z - e^{z_N T})}{(z - e^{p_1 T})(z - e^{p_2 T}) \dots (z - e^{p_N T})}$$

Impulse invariant z-transform:
Z transform of the sampled impulse response of the analog filter.

$$H(s) = \sum_{i=1}^{N} \frac{r_i}{s - p_i}$$

$$H(z) = T \sum_{i=1}^{N} \frac{r_i}{1 - e^{p_i T} z^{-1}}$$

3. Select realization structure
   - → Matched z-transform: cascade
   - → Impulse invariant z-transform: parallel

Bilinear z-transform

The basic idea is to design the digital filter by converting a continuous-time transfer function to a discrete-time equivalent.

Through bilinear z-transform, s in $H(s)$ is substituted by a function of z, namely

$$H(z) = H(s)\Big|_{s=f(z)}$$

Again, the relationship between s and z is $z = e^{sT}$

In this case, we can have

$$\ln z = sT$$

Approximation of logarithm

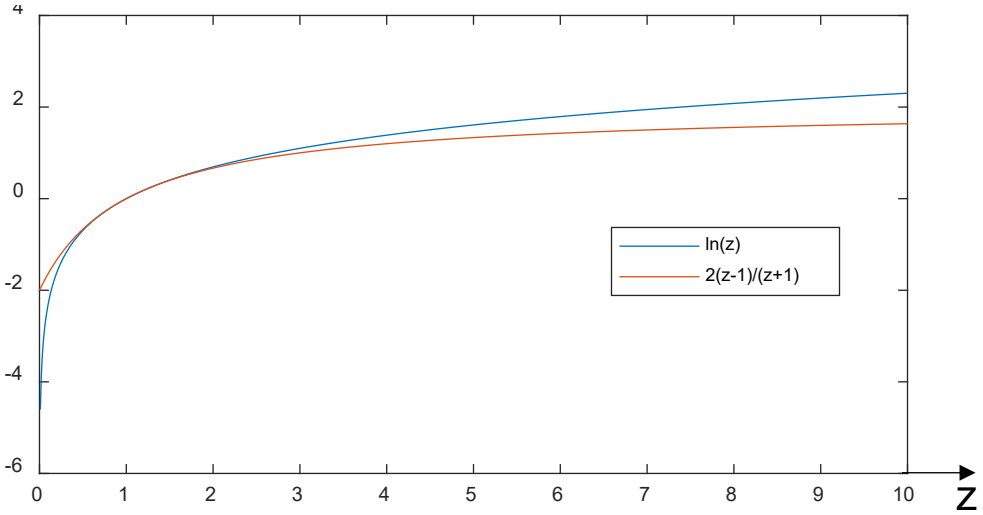
According to Taylor expansion, the natural logarithm is defined as an odd sum

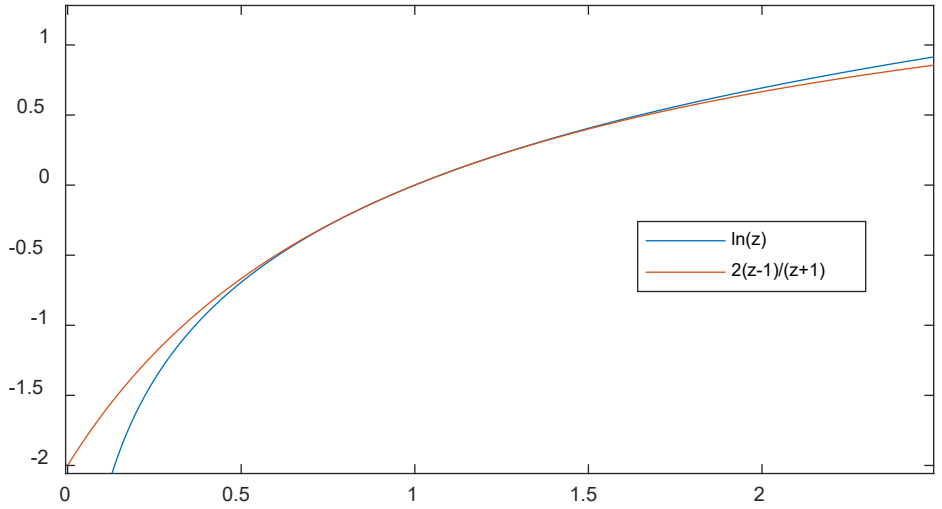
$$\ln(z) = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} (z-1)^k$$

$$\left|\frac{z-1}{z+1}\right| < 1$$

When z is about 1,

$$\ln z = 2\left[\frac{z-1}{z+1} + \frac{1}{3}\left(\frac{z-1}{z+1}\right)^3 + \frac{1}{5}\left(\frac{z-1}{z+1}\right)^5 + \ldots\right] \approx 2\frac{z-1}{z+1}$$





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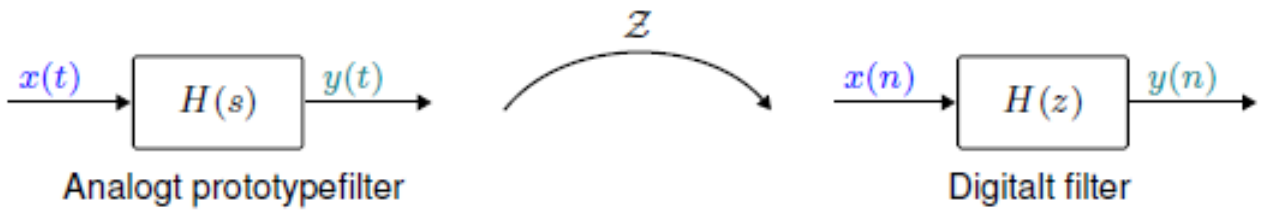
Principle

Given that

$$\ln(z) = sT \approx 2\frac{z-1}{z+1}$$

a discrete transfer function is thus obtained by bilinear z-transform as

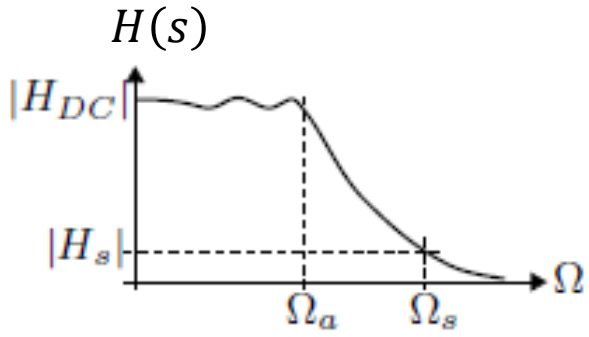
$$H(z) = H(\mathrm{s})\Big|_{s=\frac{2}{T}\frac{1-z^{-1}}{1+z^{-1}}}$$



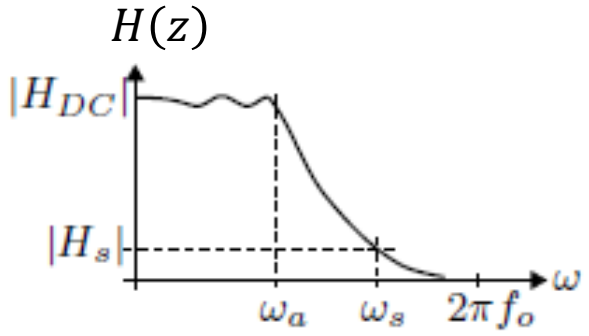
Frequency warping

Since the frequency range $0 < f < \infty$ for the **analog filter** is transformed to the frequency range $0 < f < f_o$ for the **digital filter**, the frequency axis is deformed **non-linearly**.

Analog filter



Digital filter



$$H(s) \rightarrow H(z)$$
$$z = e^{sT}$$

Q: How to choose Ω_a and Ω_s for the analog filter, so the converted digital filter can have ω_a and ω_s?

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Non-linear frequency mapping

Frequency response:
The unit circle in the z-plane corresponds to the imagine axis in the s-plane.

The relationship between **Ω** and $\boldsymbol{\omega}$ can be found by substituting $z = e^{j\omega T}$ to

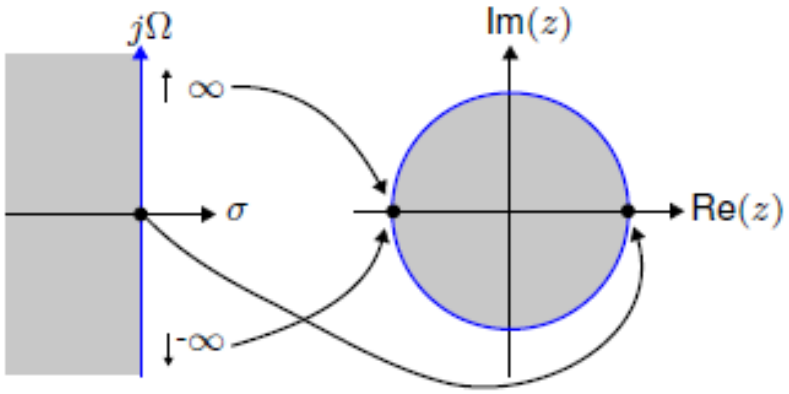
$$s = \frac{2}{T}\frac{1 - z^{-1}}{1 + z^{-1}}$$

We have

$$s = \frac{2}{T}\frac{1 - e^{-j\omega T}}{1 + e^{-j\omega T}} = \frac{2}{T}\frac{e^{j\omega T/2} - e^{-j\omega T/2}}{e^{j\omega T/2} + e^{-j\omega T/2}}$$

$$s = \frac{2}{T}\frac{2j\sin\left(\frac{\omega}{2}T\right)}{2\cos\left(\frac{\omega}{2}T\right)} = \frac{2j}{T}\tan\left(\frac{\omega}{2}T\right)$$

$$\boldsymbol{\Omega} = \frac{2}{T}\tan\left(\frac{\boldsymbol{\omega}}{2}T\right)$$

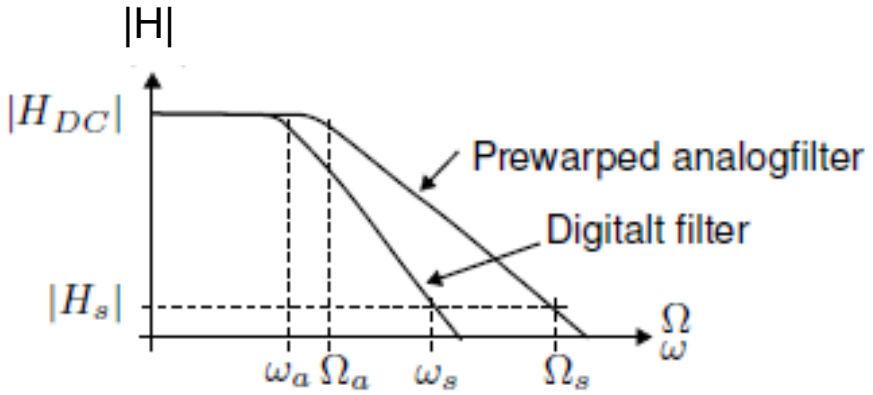


$s = j\Omega$ $\qquad$ $z = e^{j\omega T}$

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Frequency prewarping

The idea of prewarping is to adjust the analog filter's cutoff frequency Ω_a and stopband frequency Ω_s so that the converted digital filter's cutoff frequency ω_a and stopband frequency ω_s align with the desired frequencies.



$$\boldsymbol{\Omega} = \frac{2}{T}\tan\left(\frac{\boldsymbol{\omega}}{2}T\right)$$

Example

An analog 5th order Butterworth low pass filter $H(s)$ has -3 dB cutoff frequency $f_3 = \mathbf{3\ kHz}$ and -30 dB stopband frequency $f_{30} = \mathbf{6\ kHz}$.

The converted digital filter by bilinear z-transform with a sampling frequency of f_s =16 kHz has

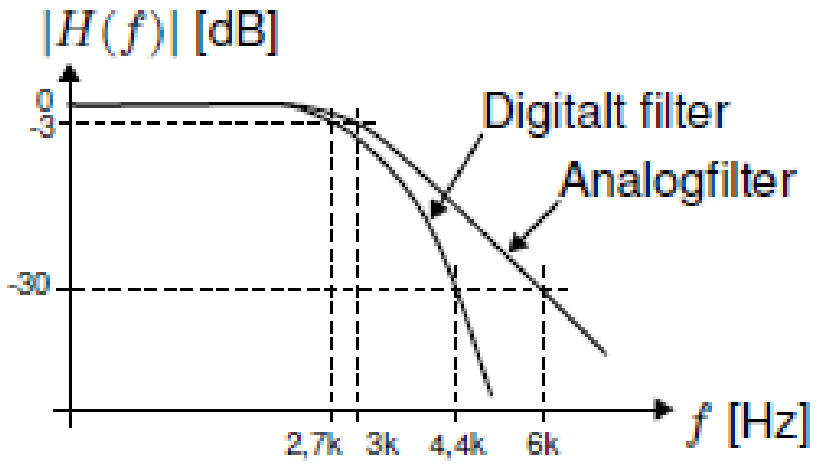
$$\omega_3 = \frac{2}{\mathrm{T}} \tan^{-1}\left(\frac{\Omega_a T}{2}\right) = 2f_s \tan^{-1}\left(\frac{2\pi f_3}{2f_s}\right) = 17.03\text{e}3 \text{ rad/s}$$ **Equal to 2.7kHz**

$$\omega_{30} = \frac{2}{\mathrm{T}} \tan^{-1}\left(\frac{\Omega_s T}{2}\right) = 2f_s \tan^{-1}\left(\frac{2\pi f_{30}}{2f_s}\right) = 27.74\text{e}3 \text{ rad/s}$$ **Equal to 4.4kHz**

Matlab:
atan()

Example

The consequence of frequency warping for the low-pass filter can be seen from the following amplitude characteristic.



Design procedure (Bilinear transform method)

1. (pre-wrap) From the given design requirements of the digital filter ω, find the analog filter's frequency Ω.

$$\boldsymbol{\Omega} = \frac{2}{T}\tan\left(\frac{\boldsymbol{\omega}}{2}T\right)$$

2. Determine the transfer function of the **analog prototype filter** $H_p(s)$

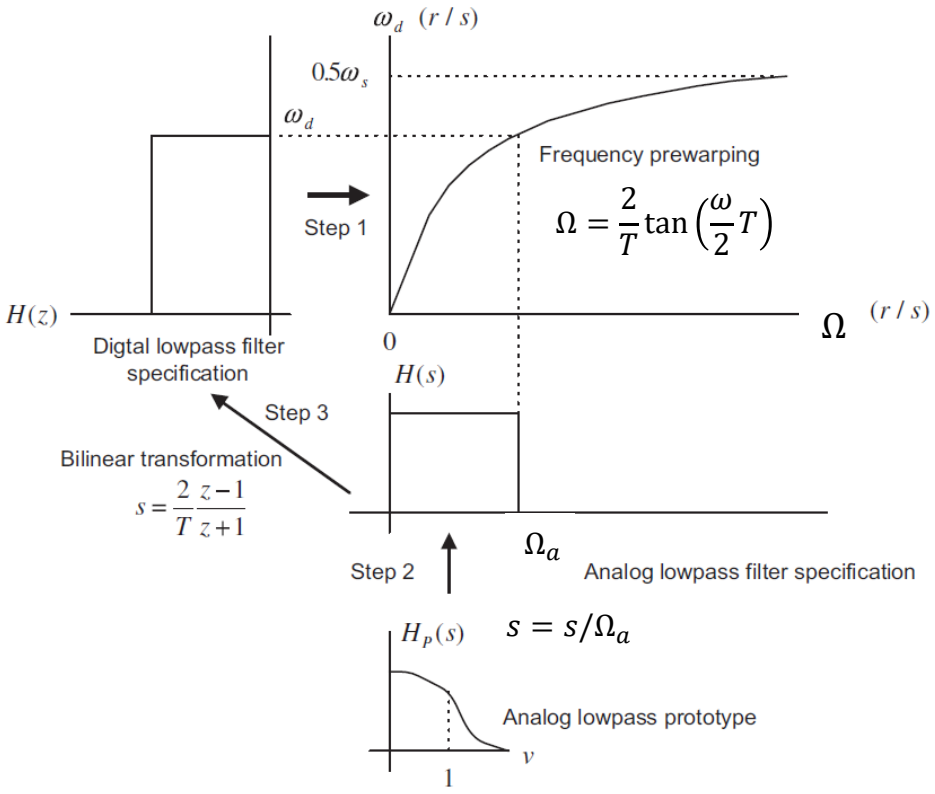
3. Denormalize

$$H(s) = H_p(s)\Big|_{s=\frac{s}{\Omega}}$$

4. Convert the analog filter to the digital filter by bilinear z-transform

$$H(z) = H(s)\Big|_{s=\frac{2}{T}\frac{1-z^{-1}}{1+z^{-1}}}$$

5. The filter is realized as a cascaded realization structure.



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Example: 1st order lowpass filter

A lowpass digital filter with a cutoff frequency of 300 Hz is to be designed with a sampling frequency of 16 kHz.
Please use bilinear z-transform method, and the **frequency-normalized** analog prototype filter should be

$$H_p(s) = \frac{1}{s+1}$$

Example

A lowpass digital filter with a cutoff frequency of 300 Hz is to be designed with a sampling frequency of 16 kHz.
Please use bilinear z-transform method, and the **frequency-normalized** analog prototype filter should be

$$H_p(s) = \frac{1}{s+1}$$

Step 1: find the pre-wrapped frequency

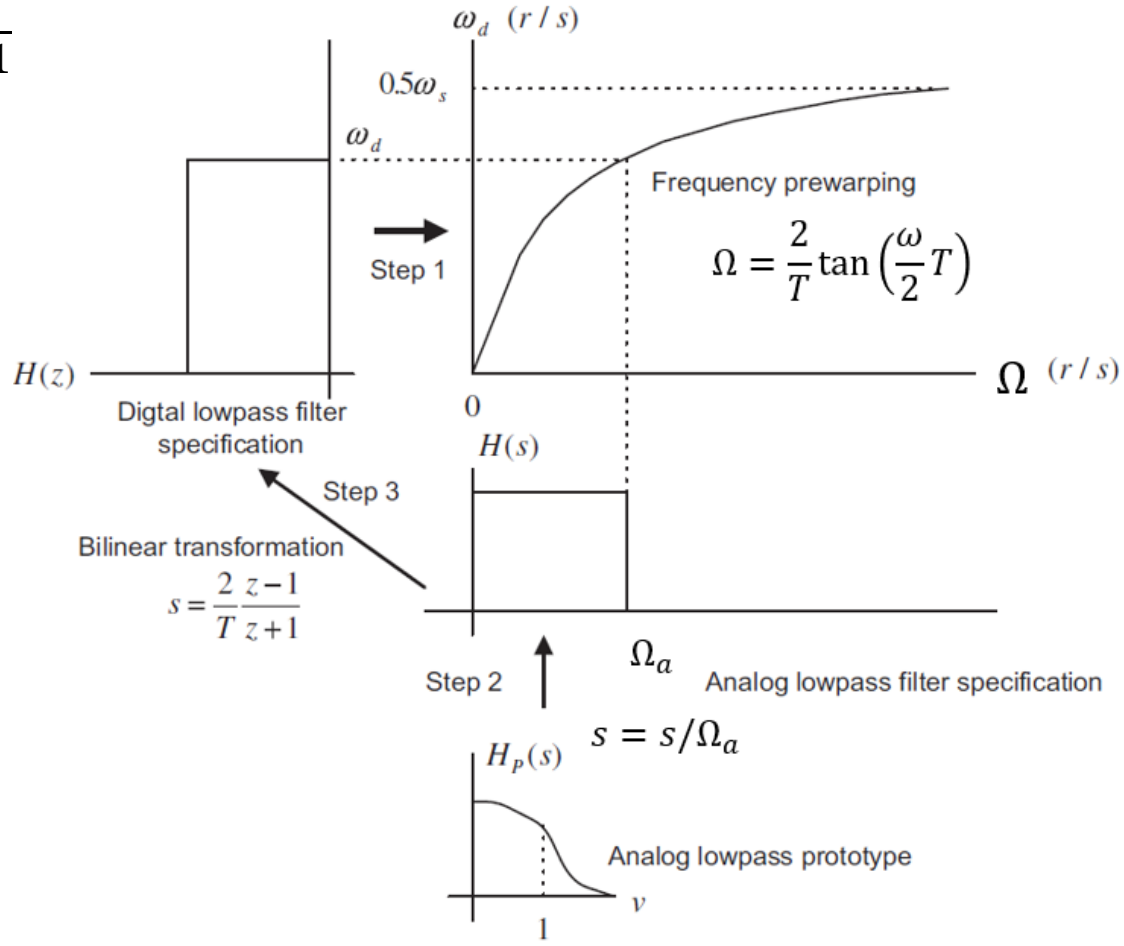
$$\Omega_a = \frac{2}{\mathrm{T}} \tan\left(\frac{\omega_a T}{2}\right) = 1887 \qquad \boxed{\omega_a = 1885 \text{ rad/s}}$$

Step 2: denormalization

$$H(s) = H_p(s)|_{s=\frac{s}{\Omega_a}} = \frac{1}{\boldsymbol{s/\Omega_a}+1} = \frac{\Omega_a}{s+\Omega_a}$$

Step 3: convert analog filter to digital filter

$$H(z) = H(s)|_{s=\frac{2}{T}\frac{1-z^{-1}}{1+z^{-1}}} = \frac{\Omega_a}{\frac{2}{T}\frac{1-z^{-1}}{1+z^{-1}}+\Omega_a} = \frac{0.056+0.056z^{-1}}{1-0.889z^{-1}}$$

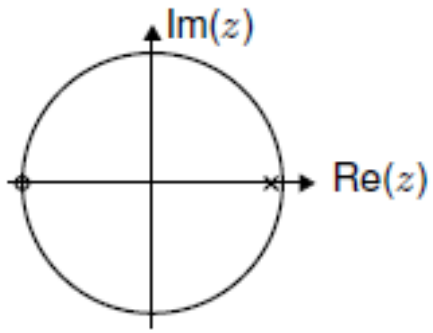


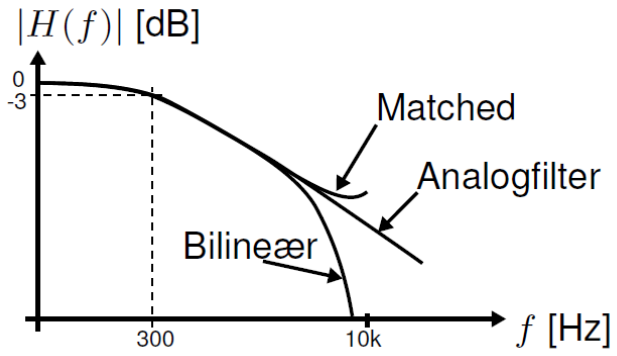
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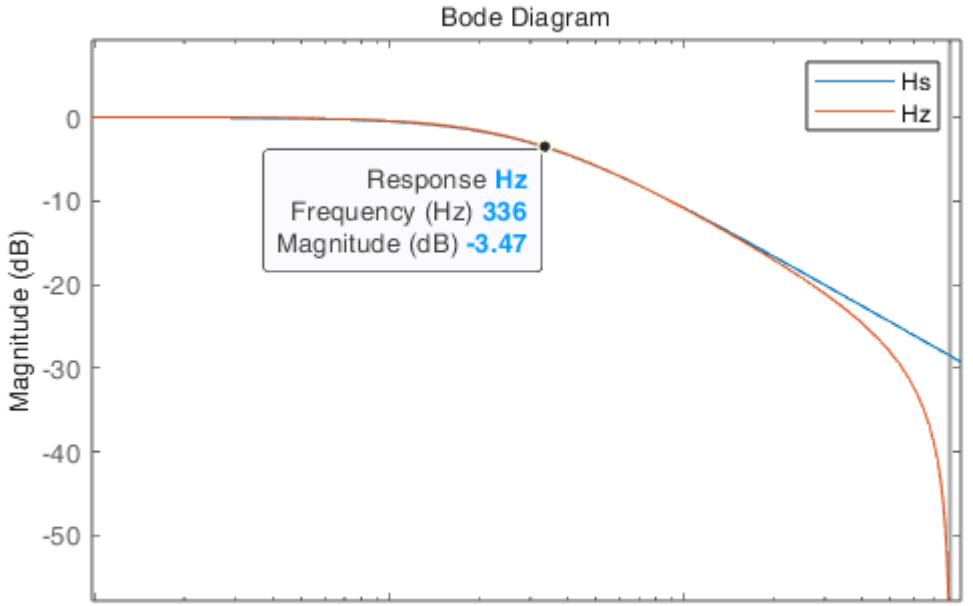
Example

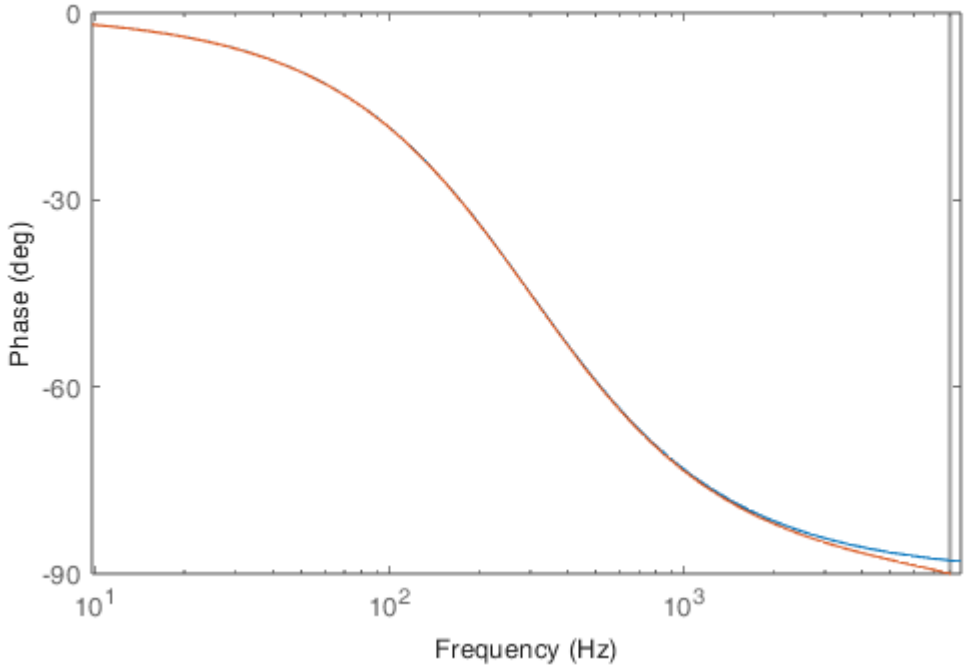
$$H(z) = \frac{0.056 + 0.056z^{-1}}{1 - 0.889z^{-1}}$$

It has a zero $z = -1$ and a pole $z = 0.889$





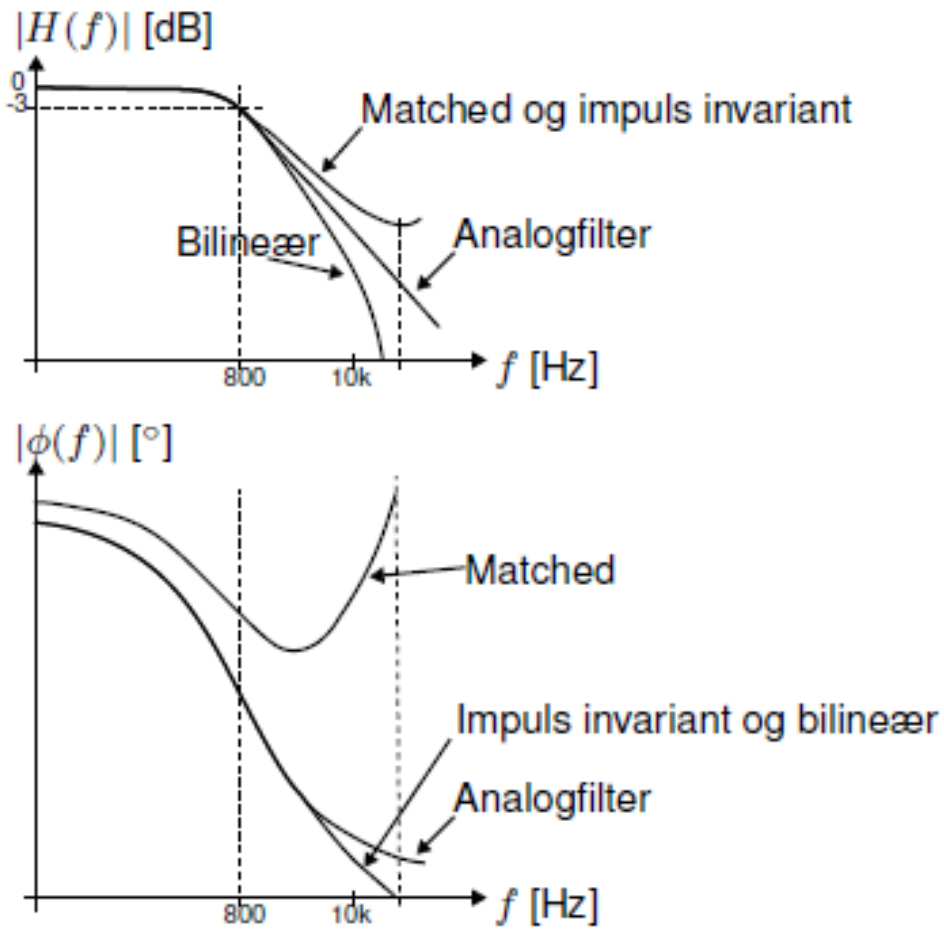




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Advantage of using bilinear z transform

You can see from the amplitude characteristic of the filter that **the aliasing error is eliminated** by bilinear z-transform.



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Matlab function

Design a digital lowpass filter with cutoff frequency 800 Hz and sampling frequency 2 kHz. Use bilinear z transform method and the **frequency-normalized prototype analog filter** is

$$H_p(s) = \frac{1}{s^2 + 1.414s + 1}$$

It is a 2nd-order Butterworth low-pass filter.

Matlab function:

H_z = c2d(Hs, 1/fs, 'Tustin');

Also, check the following function ‘bilinear’ by yourself.

→ [zd,pd,kd] = bilinear(z,p,k,fs)

→ [numd,dend] = bilinear(num,den,fs)

Matlab code:

```
fc = 800;
fs = 2000;
wc = 2*fs*tan(pi*fc/fs);  % Prewarped angular frequency

[b, a] = butter(2, wc, 's');
Hs = tf(b, a);
Hz = c2d(Hs, 1/fs, 'tustin');

options = bodeoptions;
options.FreqUnits = 'Hz';
bode(Hs, Hz, options);
grid on;
```

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What if you forget pre-warping?

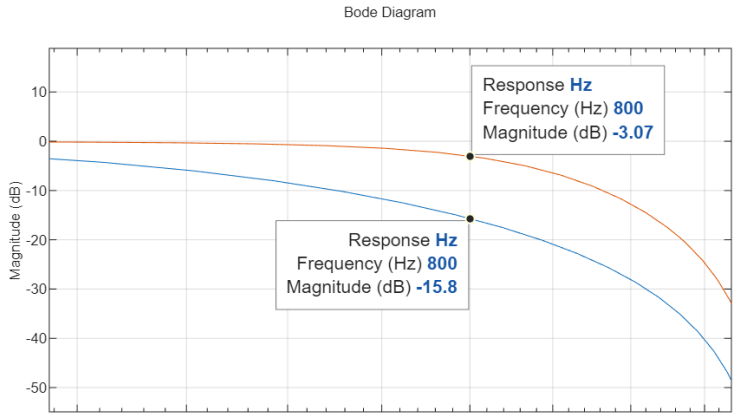
Design a digital 2nd-order Butterworth lowpass filter with cutoff frequency 800 Hz and sampling frequency **2 kHz**.
Compare the following:

Without pre-warping:

```
fc = 800;
fs = 2000;
[b, a] = butter(2, 2*pi*fc, 's');
Hs = tf(b, a);
Hz = c2d(Hs, 1/fs, 'tustin');
bode(Hz, options)
grid on
hold on
```

With pre-warping:

```
fc = 800;
fs = 2000;
wc = 2*fs*tan(pi*fc/fs);
[b, a] = butter(2, wc, 's');
Hs = tf(b, a);
Hz = c2d(Hs, 1/fs, 'tustin');
bode(Hz, options)
grid on
```



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Example: bandpass filter

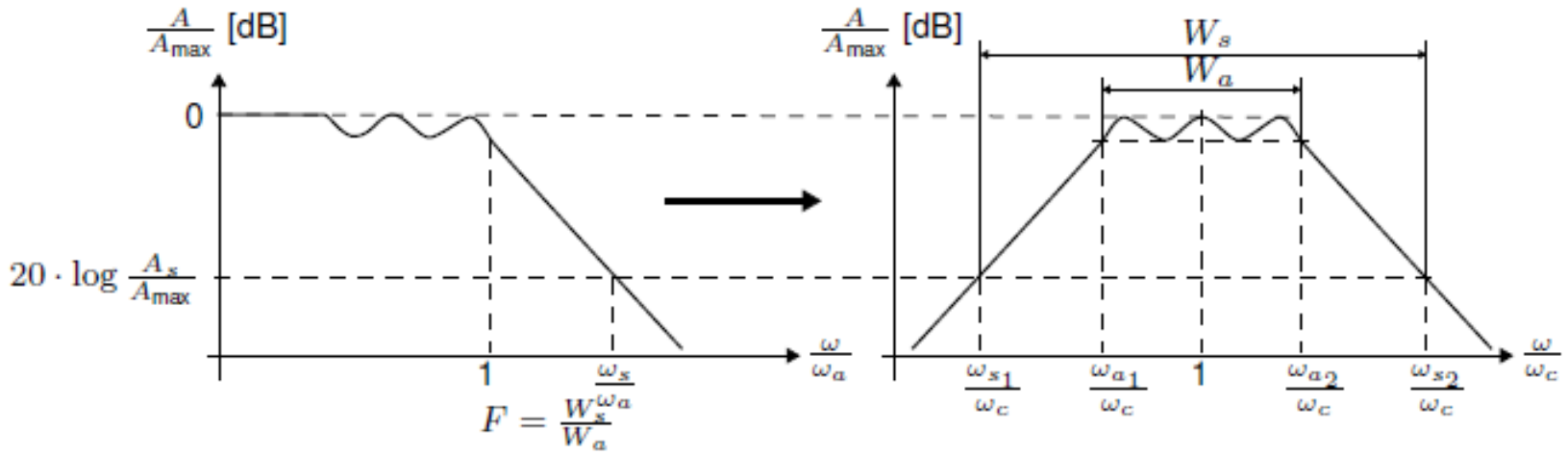
Design a digital bandpass filter with sampling frequency 100 Hz. Lower cutoff frequency 10 Hz and higher cutoff frequency 20 Hz. The **frequency-normalized** analog prototype filter should be

$$H_p(s) = \frac{1}{s+1}$$

Recap (Filter):

A low-pass filter can be transformed into a band-pass filter by

$$H_{bp}(s) = H_{lp}(\bar{s})\Big|_{\bar{s}=\frac{1}{W_a}(s+\frac{1}{s})}$$



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Example: bandpass filter

Design a digital bandpass filter with sampling frequency 100 Hz. Lower cutoff frequency 10 Hz and higher cutoff frequency 20 Hz. The **frequency-normalized** analog prototype filter should be

$$H_p(s) = \frac{1}{s+1}$$

1. (pre-wrap) From the given design requirements of the digital filter ω, find the a frequency Ω.

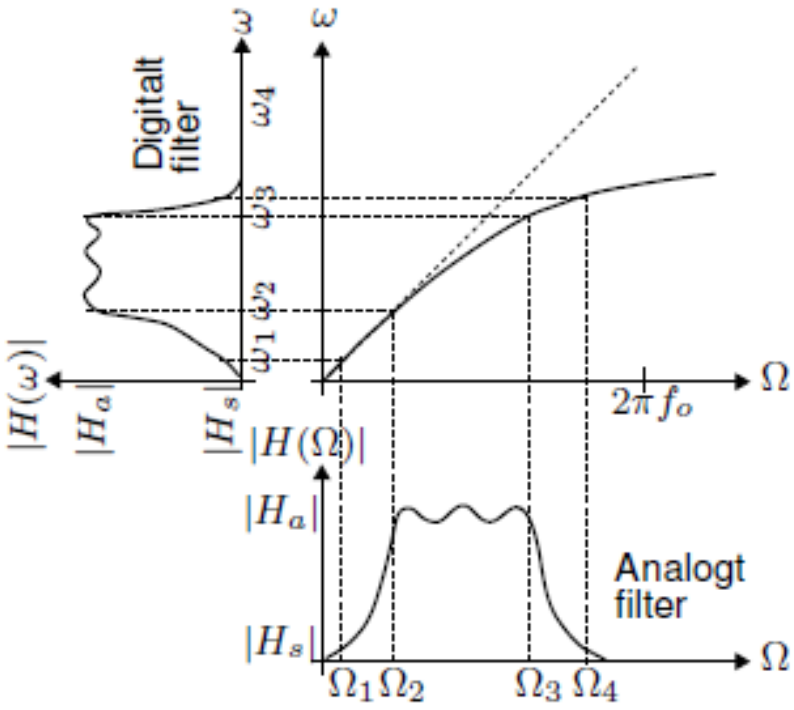
$$\boldsymbol{\Omega_L} = \frac{2}{T}\tan\left(\frac{\boldsymbol{\omega_L}}{2}T\right) = 64.984 \text{ rad/s}$$

$$\boldsymbol{\Omega_H} = \frac{2}{T}\tan\left(\frac{\boldsymbol{\omega_H}}{2}T\right) = 145.309 \text{ rad/s}$$

Calculate the central frequency

$$\boldsymbol{\Omega_C} = \sqrt{\boldsymbol{\Omega_L}\boldsymbol{\Omega_H}} = 97.174 \text{ rad/s}$$

$$W_a = \frac{\boldsymbol{\Omega_H} - \boldsymbol{\Omega_L}}{\boldsymbol{\Omega_C}} = 0.8266$$



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Example: bandpass filter

Design a digital bandpass filter with sampling frequency 100 Hz. Lower cutoff frequency 10 Hz and higher cutoff frequency 20 Hz. The **frequency-normalized** analog prototype filter should be

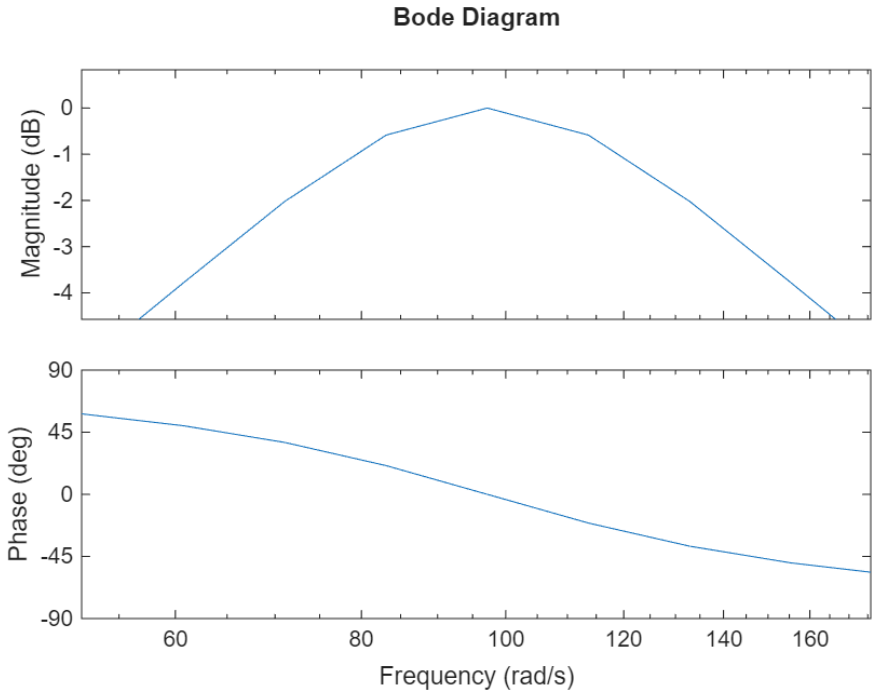
$$H_p(s) = \frac{1}{s+1}$$

2. Determine the transfer function of the analog filter

$$H_{bp}(s) = H_{lp}(\bar{s})\Big|_{\bar{s}=\frac{1}{W_a}(s+\frac{1}{s})} = \frac{0.8266s}{s^2+0.8266s+1}$$

Denormalize

$$H_{BP}(s) = H_{bp}(s)\Big|_{s=\frac{s}{\Omega_c}} = \frac{80.324s}{s^2+80.324s+9442.8}$$



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Example: bandpass filter

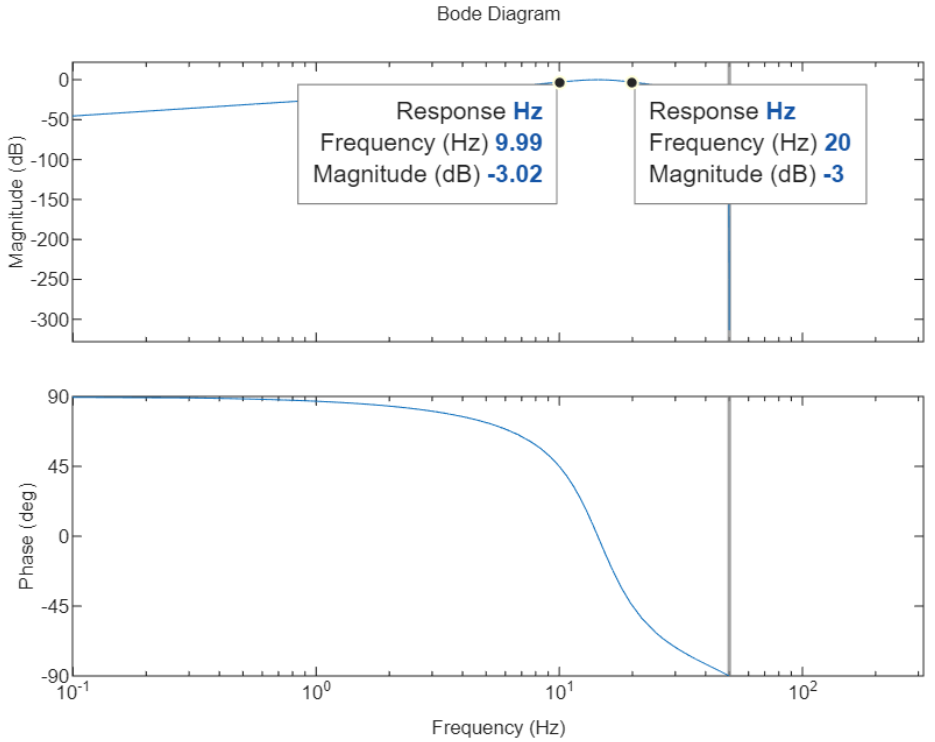
Design a digital bandpass filter with sampling frequency 100 Hz. Lower cutoff frequency 10 Hz and higher cutoff frequency 20 Hz. The **frequency-normalized** analog prototype filter should be

$$H_p(s) = \frac{1}{s+1}$$

3. Convert the analog filter to the digital filter by bilinear z-transform

$$H(z) = H(s)\Big|_{s=\frac{2}{T}\frac{1-z^{-1}}{1+z^{-1}}} = \frac{0.2452 - 0.2452z^{-2}}{1 - 0.9329z^{-1} + 0.5095z^{-2}}$$

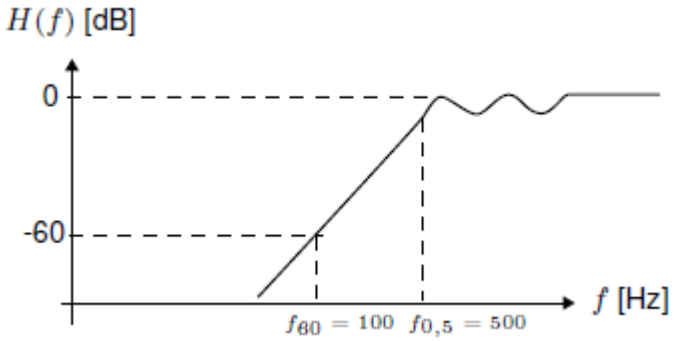
```
Hs = tf([80.324,0],[1, 80.324, 9442.8]);
Hz = c2d(Hs, 1/fs, 'tustin');
Bode(Hz)
```



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Exercise: Design a highpass filter

Design a 0.5 dB Chebyshev highpass filter with cutoff frequency 500Hz, sampling frequency 8kHz.



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Exercise

8.15. Design a second-order bandstop digital Chebyshev filter with the following specifications:

Center frequency of 2.5 kHz

Bandwidth of 200 Hz

1 dB stopband ripple

Sampling frequency of 8,000 Hz.

a. Determine the transfer function and difference equation.

b. Use MATLAB to plot the magnitude and phase frequency responses.

Use 3 methods as follows. Bode Plot of them, together with the H(s), and compare:

- Matched z-transform
- Impulse invariant transform
- Bilinear z transform